

Splashing the Surface of Research: A Study of Koli Abstracts

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ABSTRACT

Water is wet. The Pope is Catholic. In this poster, we explore what else isn't new to the Koli Calling research community. We hypothesize that the introductory sentences of abstracts reflect the community's shared assumptions. Using a mixed-methods approach, we analyze all the abstracts published in Koli proceedings to date. The results suggest, among other things, that the community believes programming to be difficult and online education to be common. For deeper insights, we would have needed to actually read the papers.

CCS CONCEPTS

• **Social and professional topics** → Computing education.

KEYWORDS

abstracts, research community, shared assumptions, Koli, pool poster

ACM Reference Format:

Juha Sorva. 2019. Splashing the Surface of Research: A Study of Koli Abstracts. In *19th Koli Calling International Conference on Computing Education Research (Koli Calling '19)*, November 21–24, 2019, Koli, Finland. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3364510.3366148>

1 BACKGROUND AND GOALS

Many writing guides suggest that an article abstract should first provide context and identify a gap in existing research, then summarize the article's contribution. To be sure, many abstracts follow this pattern: there's a *preamble*, a *"but,"* and a *payload* (Fig. 1).

Preambles are not universally loved. According to a modern English phrase, "everything before the 'but' is bullshit"—and bullshit is, as Frankfurt [2] has eloquently argued, "a greater enemy of the truth than lies are." In a less refined but more pertinent line of criticism, the hackneyed opening statements of research reports have drawn occasional comment at Koli Calling. One wag reportedly planned to begin all their articles with "[*Paper topic*] is very important" while another held faith with "*Programming is difficult.*"

In this poster, we build on the observation that many articles start with claims that are uncontroversial to the point of banality. We take up the hypothesis that those claims reflect the shared assumptions of the research community that writes and reviews the articles; we set out to explore that socially constructed reality. More specifically, we seek to find out which claims are sufficiently generic and unobjectionable to start a Koli abstract with. As our work is

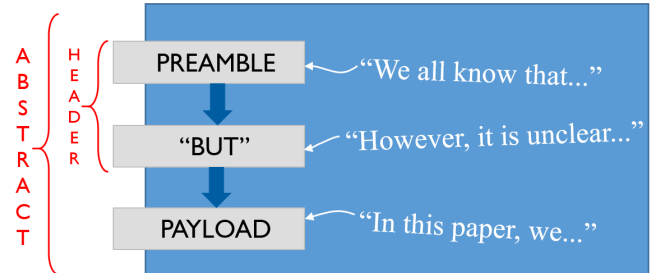


Figure 1: Terms used in this poster. The prototypical abstract consists of a preamble, a “but,” and a payload. Everything that comes before the payload forms the header.

both a methodological exploration and a germinal study in this area, we complement our main findings by reporting descriptive statistics that quantify the data we collected and the abstract-writing habits of the community.

2 RELATED WORK

Several metastudies of Koli articles have been published, most recently in 2016 [7], but none are as shallow as ours or have the same focus. A session with Google Search did not turn up comparable studies in other research communities, either.

In any case, it may be that the work closest to ours in spirit gets published in unconventional forums [5].

3 METHOD AND RESULTS

We collected all Koli Calling abstracts from 2001 to 2018, ignoring keynotes, workshops, and doctoral consortia as well as 28 articles (mostly from 2001 and 2002) that had no abstract. This left us with a pool of 406 articles: 82 posters, 137 short papers (including discussion papers), and 187 long papers (including system papers).

3.1 Overall statistics

We compared the abstract lengths of posters ($M = 97.7$ words, $SD = 45.2$, $min = 25$, $max = 285$), short papers ($M = 119.5$, $SD = 45.0$, $min = 45$, $max = 273$), and long papers ($M = 153.0$, $SD = 50.7$, $min = 47$, $max = 327$). A one-way ANOVA yielded an infinitesimal p-value ($F(2, 403) = 43.79$, $p = 0.00$) and a post-hoc Tukey test indicated pairwise differences between all group pairings.

We sectioned each abstract into a header and a payload as in Fig. 1. The payload begins where the authors describe their article's purpose; we refer to beginning the payload as *getting to the point*. Although we further divided the header into a preamble and a “but,” we will ignore that distinction here; our analysis covers all the text that appears before the authors get to the point.

For each abstract, we computed a *distance-to-point* statistic (DtP), which is the number of words before the authors get to the point divided by the abstract's total length. Posters had an average DtP of 0.31 ($SD = 0.27$, $max = 0.82$), short papers 0.26 ($SD = 0.24$, $max =$

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Koli Calling '19, November 21–24, 2019, Koli, Finland

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ACM ISBN 978-1-4503-7715-7/19/11...\$15.00

<https://doi.org/10.1145/3364510.3366148>

0.89), and long papers 0.24 ($SD = 0.22$, $max = 0.85$). The substantial standard deviations are in part because some articles had an empty header. To quantify this, we computed a *straight-to-point aggregate* (SPA) for all the articles, defined as the proportion of abstracts whose DtP equals zero. The SPA of our data was 32.8%.

3.2 Shared assumptions of the community

We read the headers of all the abstracts, seeking to identify the main statements made in each sentence. From the resulting list of 820 statements, we discarded 92 that defined a term, system, or overall project rather than making a proposition about them.

The vast majority of the other statements claimed that something was either important, difficult, common, beneficial, time consuming, neglected, or unknown. Within those broad categories, we identified recurrent themes, which we call *gists*; each gist is our attempt to distill the meaning of the statements within it. We discarded all gists that did not feature statements from at least two disjoint sets of authors.

The gists in the box below represent our primary findings on the Koli community's shared assumptions. Each of these eleven gists paraphrases at least four—and sometimes more than a dozen—statements that are similar to each other.

Learning to program is hard. Computational thinking has become big in K-12 education. So has programming. On-line education is now common. Computer science courses have high drop-out rates. Automatic assessment is common in programming courses. It is unclear how to teach programming well. Feedback is essential to learning. Teachers should know student misconceptions. Reflection is good for learning. Automatic assessment tools provide feedback.

Space precludes a full list of the other gists, but here is a selection:

Teamwork skills are valuable. Women are underrepresented in computer science. Success in learning to program requires practice. Student motivation is important. Automatic assessment reduces teachers' workload. Computers are part of our daily life. Games are a popular form of recreation. Social media is used by a lot of people these days. Plagiarism is a common problem.

We did not attempt to check inter-rater reliability since our colleagues no doubt had better things to do. (We were on holiday.)

4 DISCUSSION

Our results show that there is great variation in reaching the point. Roughly a third of Koli papers get straight to the point but some take more than eighty percent of the abstract to do so.

Most of the claims that Koli authors make before they get to the point appear quite innocuous. This anecdotal observation on our part is hardly conclusive evidence that our working hypothesis holds water—i.e., that abstract beginnings reflect shared assumptions. Nevertheless, it does lend a modicum of face validity to our approach.

Overall, we feel that our gists capture a set of impressively uncontroversial statements. The next question is what to do with them; this is something for the community to decide. We will comment on the matter only briefly.

One option is to avoid these humdrum statements and either get straight to the point or open with something more thought-provoking. The first sentence of an article—scientific papers included—may decide if the article sinks or swims, so it should attract the reader's attention. Having now identified some recurrent opening statements in Koli articles, perhaps we can take them as given and no longer use them to start our articles?

Another thing we might do with the gists is subject them to critical scrutiny. For inspiration, we only need to look at “water is wet,” a statement so obvious that it is routinely employed by English-speakers to dismiss news that isn't newsworthy. Even so, opinions on water's wetness are in fact sharply divided [1]. Ultimately, what counts as wet depends on what one means by “wet”: the ability of liquid to stick to surfaces or something more experiential [6].

Consider now one of our gists: that programming is hard. This common claim, too, has been called into question [4]: children do learn to program, after all, and CS1 teachers expect rather a lot in a short time. On the other hand, what is taught to children is only a limited form of programming, and programming is harder than spoken language and other things that come naturally to humans [3]. Clearly, whether programming is hard or not depends on what one means by “programming” and “hard.”

The Pope is, nevertheless, very likely to be Catholic.

5 CONCLUSION

The beginnings of articles are a wellspring of unsurprising information. In them, we may have uncovered some of the shared assumptions of the Koli Calling research community. Be that as it may, we feel vindicated in our decision to explore Koli abstracts, which had been previously overlooked as a source of bromides. The prospect of applying this approach to other communities makes our head swim.

Actually understanding the point of the papers was, of course, beyond the scope of this study. Who knows what we might have found if we had more than dipped our toes into the literature?

6 ACKNOWLEDGEMENTS

We thank the Koli-goers who dreamed up an evening track of “pool posters” to be presented at the hotel spa. It is that standard of quality that this work aspires to.

This poster is dedicated to the lasting legacy of Keeko Valaskala.

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